

Classification of Research

Week 2-3 Lecture

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Agenda

Recap of the Last Lectures

Basic Terminologies used in Research

Classification of Research

Types of Research according to purpose

Types of Research according to use

Types of Research according to time

Types of Research according to data collection

Introducing a research process

Recap of the Lecture



Classification of Research

Classification of Research



Dimension I- According to Purpose



Dimension II-According to Use



Dimension III-According to Time



Dimension IV-According to Data Collection

Dimension I – Purpose of doing research

- If we ask someone why he or she is conducting a study, we might get a range of responses: “My boss told me to do”; “It was a class assignment”; “I was curious.” There are almost as many reasons to do research as there are researches.
- Yet the purposes of research may be organized into **three groups** based on what the researcher is trying to accomplish – **explore a new topic, describe a social phenomenon, or explain why something occurs.**
- Studies may have multiple purposes (e.g. both to explore and to describe) but one purpose usually dominates.

Dimension I – Purpose of doing research

- Hence, as per dimension-I, there are three types of research:

1.1 Exploratory or Formulative Research

1.2 Descriptive Research

1.3 Explanatory Research

1.1 Exploratory or Formulative Research

- You may be **exploring a new topic** or issue in order to learn about it .
- The researcher's goal is to formulate more precise questions that future research can answer.
- Exploratory research may be the first stage in a sequence of studies.
- A researcher may need to know enough to design and execute a second, more systematic and extensive study.
- The findings discovered through exploratory research would encourage the researchers to emphasize learning more about the particulars of the findings in subsequent conclusive studies.
- For exploratory research, the researcher may use different sources for getting information like (1) experience surveys, (2) secondary data analysis, (3) case studies, and (4) pilot studies.

1.1 Exploratory / Formulative Research

•**Goal:** To explore a new topic, issue, or phenomenon.

•**Focus:** Formulating clearer questions for future research.

•**Methods:**

- Experience surveys
- Secondary data analysis
- Case studies
- Pilot studies

•**Example:**

•Initial investigation into a new social media trend's impact on youth.

•Investigating potential uses of quantum computing in healthcare

•Exploring early user reactions to a new AR/VR headset

•Preliminary study on blockchain applications in supply chain

•Initial investigation of AI-generated art's impact on traditional artists

•Exploring how Gen Z uses social media for political engagement

Example of Exploratory or Formulative Research

1. The favorite crime investigation TV programs (e.g., *Crime Patrol*, *Arjun*, *Savdhaan India*, etc) give a pretty good example of the research design. These shows typically start with a crime that needs to be investigated. The initial step is to look for hints which can help establish what has happened (exploratory). The clues found in the exploratory phase of the research usually point in the direction of a specific hypothesis or explanation of the events which happened, and investigators start focusing their efforts in this direction, performing interviews with witnesses and suspects (descriptive).

1.2 Descriptive Research

- *Descriptive research* presents a picture of the specific details of a situation, social setting, or relationship.
- Labor Force Surveys, Population Census, and Educational Census are examples of such research.
- Example 1: A school district that wants to understand if students will access online lessons rather than textbooks.
- Example 2: Market researchers that want to observe habits of consumers.

1.2 Descriptive Research

- **Goal:** To describe a situation, setting, behavior, or phenomenon in detail.
- **Focus:** “What is happening?”
- **Methods:**
 - Surveys
 - Observations
 - Census data

Example: Survey on student preferences for online vs. textbook learning

Census data analysis of internet penetration across rural areas

Survey of remote work patterns post-pandemic

Documentation of e-commerce shopping behaviors during holiday seasons

Study describing social media usage patterns by age group

Analysis of smartphone brand preferences across different income brackets

1.3 Explanatory Research

- When we encounter an issue that is already known and have a description of it, we might begin to wonder *why* things are the way they are.
- The desire to know “why”, to explain, is the purpose of *explanatory research*.
- Explanatory research looks for causes and reasons.
- For example, a descriptive research may discover that 10 percent of the parents abuse their children, whereas the explanatory researcher is more interested in learning *why* parents abuse their children.

1.3. Explanatory Research

- **Goal:** To explain causes, reasons, or relationships.
- **Focus:** “Why is this happening?”
- **Methods:**
 - Experiments
 - Longitudinal studies
 - Statistical analysis

Example: Investigating *why* parents abuse their children.

Why certain algorithms outperform others in image recognition

Investigating causes of employee burnout in tech companies

Why some social media platforms retain users better than others

Determining factors influencing cryptocurrency adoption

Why certain cybersecurity threats are more prevalent in specific industries

Dimension II – The use of Research

- Some researchers focus on using research to advance general knowledge, whereas others use it to solve specific problems.
- Those who seek an understanding of the fundamental nature of social reality are engaged in **basic research** (also called academic research or pure research or fundamental research).
- **Applied research**ers, by contrast, primarily want to apply and tailor knowledge to address a specific practical issue. They want to answer a policy question or solve a pressing social and economic problem.

2.1. Basic (Pure / Fundamental) Research

Goal: To expand knowledge, develop theories, and understand fundamentals.

No immediate practical application.

Examples:

1. Studying neural network architectures for theoretical understanding
2. Research on fundamental properties of quantum entanglement
3. Investigating mathematical foundations of encryption algorithms
4. Studying cellular automata theory
5. Research on fundamental principles of human-computer interaction

2.2. Applied Research

Goal: To solve practical, real-world problems.

Directly applicable to policy, products, or practices.

Examples:

1. Developing a new fraud detection system for banks
2. Creating a traffic optimization algorithm for smart cities
3. Designing a machine learning model for early disease detection
4. Developing energy-efficient algorithms for IoT devices
5. Creating personalized learning recommendation systems

Basic versus Applied Research

BASIS FOR COMPARISON	BASIC RESEARCH	APPLIED RESEARCH
Meaning	Basic Research refers to the study that is aimed at expanding the existing base of scientific knowledge.	Applied Research is the research that is designed to solve specific practical problems or answer certain questions.
Nature	Theoretical	Practical
Utility	Universal	Limited
Concerned with	Developing scientific knowledge and predictions	Development of technology and technique
Goal	To add some knowledge to the existing one.	To find out solution for the problem at hand.

Example of Basic Research

- How did the universe begin?
- What are protons, neutrons, and electrons composed of?
- How do slime molds reproduce?
- What is the specific genetic code of the fruit fly?

Example of Applied Research

- improve agricultural crop production
- treat or cure a specific disease
- improve the energy efficiency of homes, offices, or modes of transportation

Dimension III – The Time Dimension in Research

- Another dimension of research is the treatment of time.
- Some studies give us a snapshot of a single, fixed time point and allow us to analyze it in detail.
- Other studies provide a moving picture that let us follow events, people, or sale of products over a period of time.
- In this way from the angle of time research could be divided into two broad types:
 - Cross Sectional Research
 - Longitudinal Research

Cross Sectional Research

- *Cross-Sectional research*, researchers observe at one point in time.
- For example, The use of CAM (Complementary and Alternative Medicine) among pregnant women

3.1. Cross-Sectional Research

Snapshot of a phenomenon at **one point in time**.

Examples:

1. 2024 survey of AI adoption across different industries
2. Snapshot analysis of coding bootcamp graduates' employment status
3. One-time assessment of cybersecurity awareness among employees
4. Current analysis of cloud computing service usage patterns
5. Single-point study of mobile app user satisfaction ratings
6. Survey on CAM (Complementary and Alternative Medicine) use among pregnant women in 2024.

Longitudinal Research

- Researchers using longitudinal research examine features of people or other units at more than one time.
- For example: Time Series Research or several clinical trials in medicine

3.2. Longitudinal Research

Study conducted **over a period of time**.

Tracks changes, trends, or developments.

Examples:

1. Tracking career progression of CS graduates over 15 years
2. Monitoring climate change effects on agricultural software adoption (2010-2024)
3. Yearly assessment of digital literacy improvements in rural communities
4. Long-term study of programming language popularity trends
5. 10-year follow-up of startup success rates after accelerator programs
6. Following a cohort of CS graduates' career paths over 10 years.

Dimension IV – The Data Collection

Dimension in Research

- Every researcher collects data using one or more techniques.
- The techniques may be grouped into two categories: *quantitative* and *qualitative*.
- **Example:** Effect of web-browsing interfaces in web-based instruction: A quantitative study
- In quantitative research, the data is collected in the form of numbers. For instance, surveys.
- In qualitative research, the data is collected in the form of words or pictures. For example, interviews.
- **Example:** Team versus individually undergraduate FYP: a qualitative study of student experiences and preferences

4.1. Quantitative Research

Data Type: Numbers, statistics.

Methods:

- Surveys with closed-ended questions

- Experiments

- Statistical analysis

Examples:

- 1.A/B testing of website conversion rates

- 2.Statistical analysis of app download numbers across regions

- 3.Performance benchmarking of different database systems

- 4.Survey with Likert-scale questions about software usability

- 5.Analysis of server log data to identify usage patterns

- 6.Measuring student test scores before and after a new teaching method.

4.2. Qualitative Research

Data Type: Words, images, observations.

Methods:

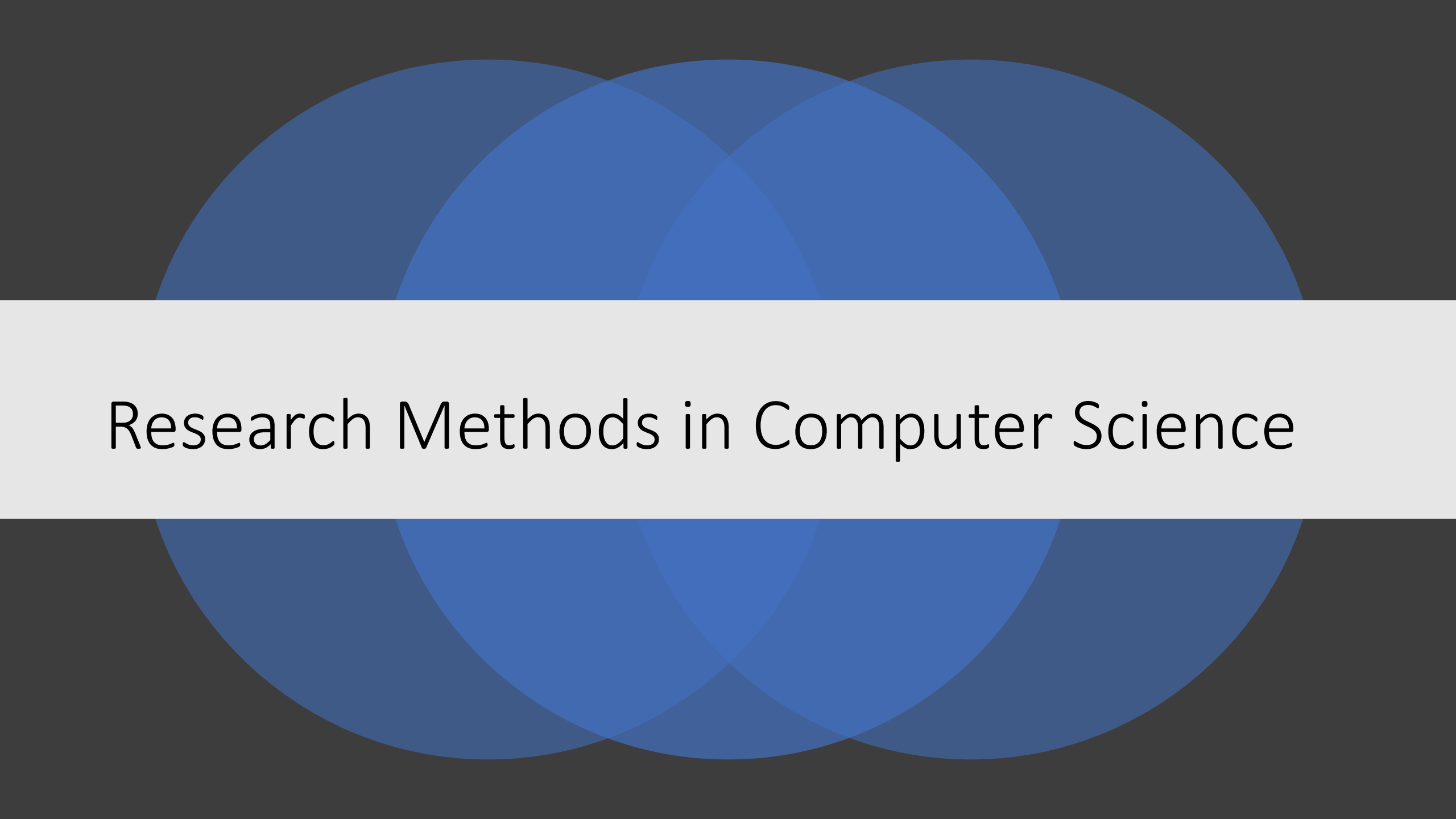
- Interviews

- Focus groups

- Content analysis

Examples:

1. In-depth interviews with cybersecurity experts
2. Focus groups discussing user experience of a new interface
3. Ethnographic study of programmers' collaboration patterns
4. Content analysis of developer forum discussions
5. Case study of successful digital transformation in a company
6. Interviewing students about their learning experiences.



Research Methods in Computer Science

Research Methods in Computer Science

1. Implementation-driven Research
2. Mathematical Proof Techniques
3. Quantitative Research
4. Qualitative Research
5. Mixed Method Research

Implementation-driven Research

- This approach progresses by iteratively building better and better systems.

Mathematical Proof Techniques

- This approach uses formal proofs to reason about the validity of a hypothesis given some evidence.
- For example, mathematical reasoning can be used to demonstrate that an algorithm can cover all possible input cases.

Qualitative Research

- An array of interpretive techniques which seek to describe, decode, translate in terms of meaning not the frequency.
- Aims to achieve an in-depth understanding of a situation.

Quantitative Research

- Attempts precise measurement of some behavior, knowledge, opinion or attitude.
- Operationalization of variables

Mixed Method Research

- A combination of Qualitative and Quantitative

4.3. Mixed Methods Research

Combines quantitative and qualitative approaches.

Provides both numerical trends and in-depth understanding.

Examples:

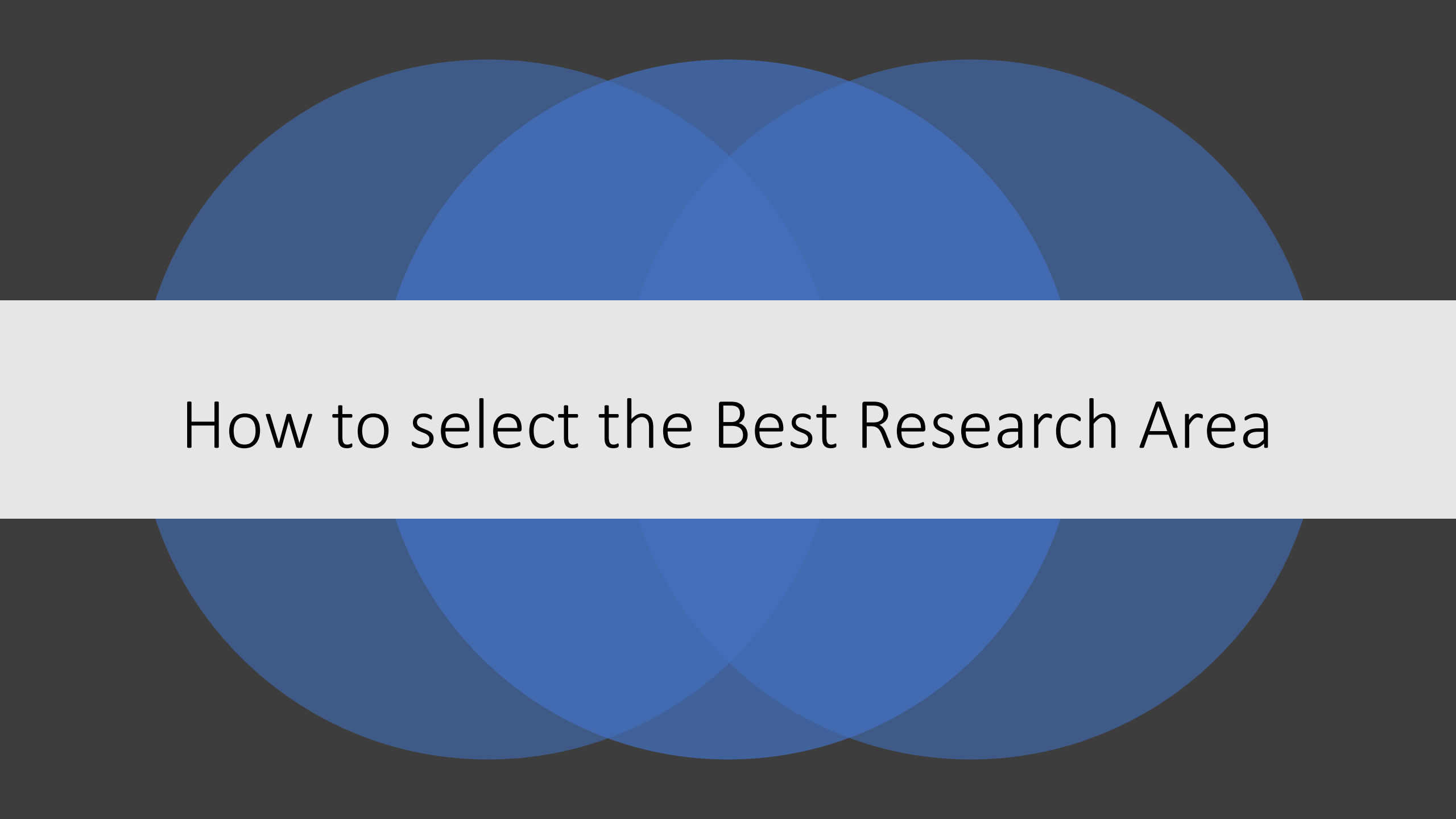
1. Survey (quantitative) + Follow-up interviews (qualitative) about remote work tools
2. System performance metrics + User diaries about their experiences
3. Web analytics + User testing observations
4. Pre/post-test scores + Participant reflections
5. Social media sentiment analysis + Focus group discussions
6. Survey (quantitative) + Follow-up interviews (qualitative).



Research Process

Research Process





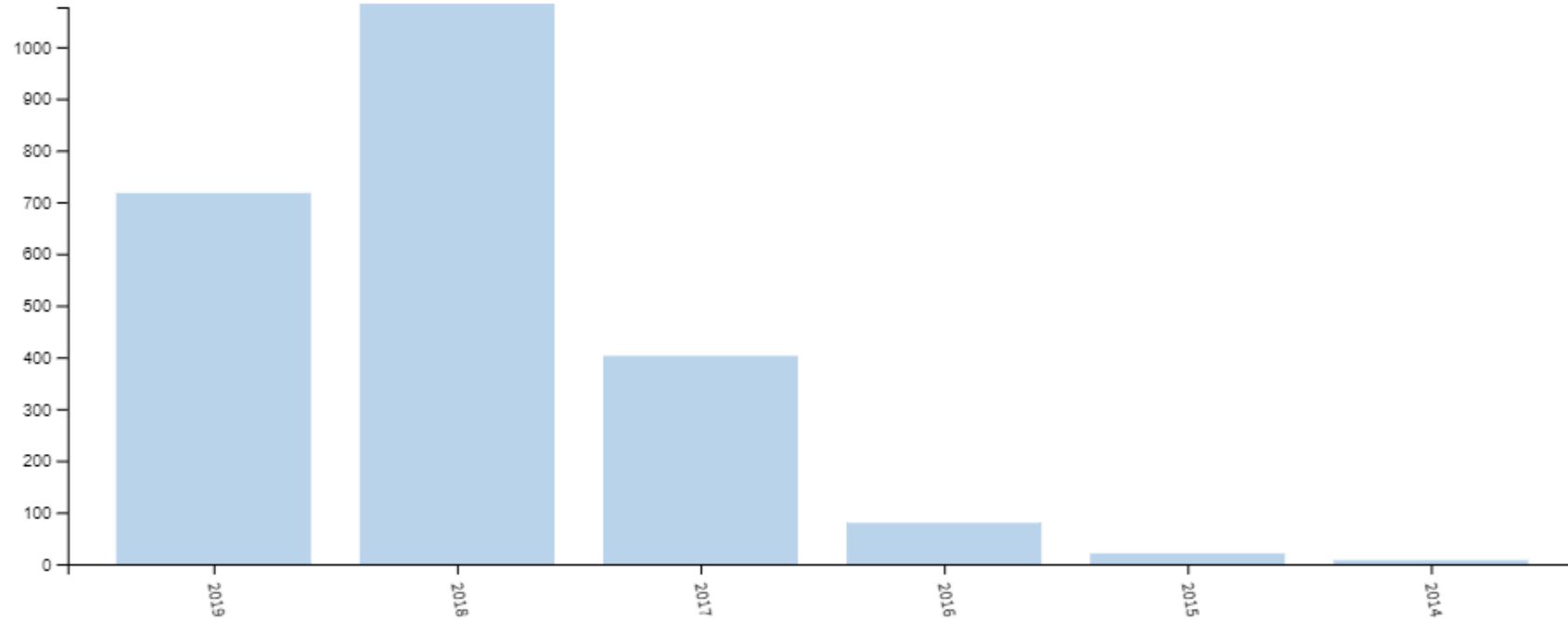
How to select the Best Research Area



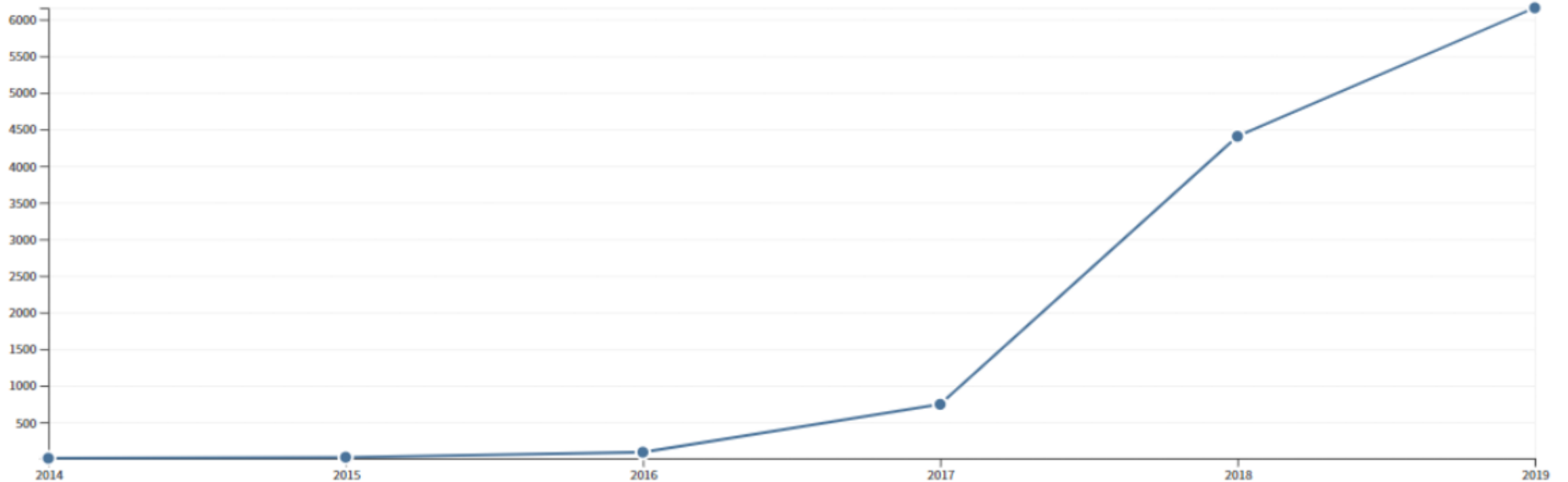
Research Areas in the field of CS

- Machine Learning
- Deep Learning
- Big Data Analytics
- Blockchain
- Cyber Security
- Cloud Computing
- Fog Computing
- Network Security
- Software Defined Network (SDN)
- And several others.....

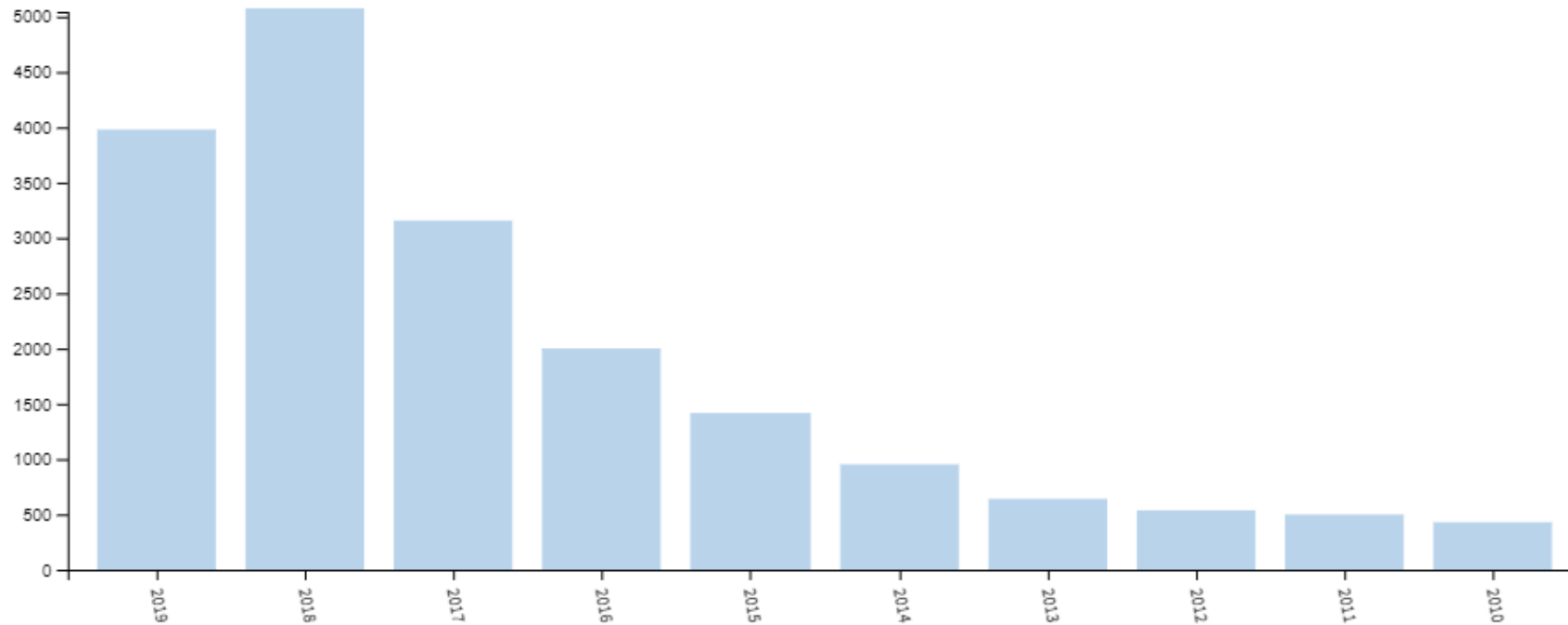
Blockchain- Publication Analysis



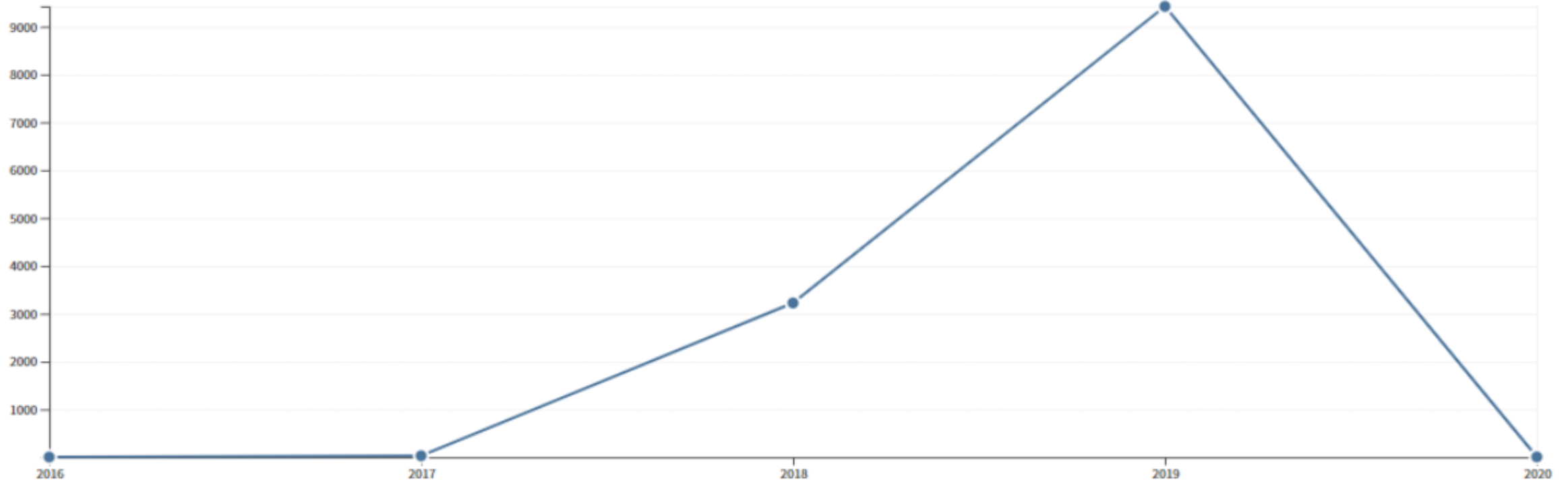
Blockchain- Citations Analysis

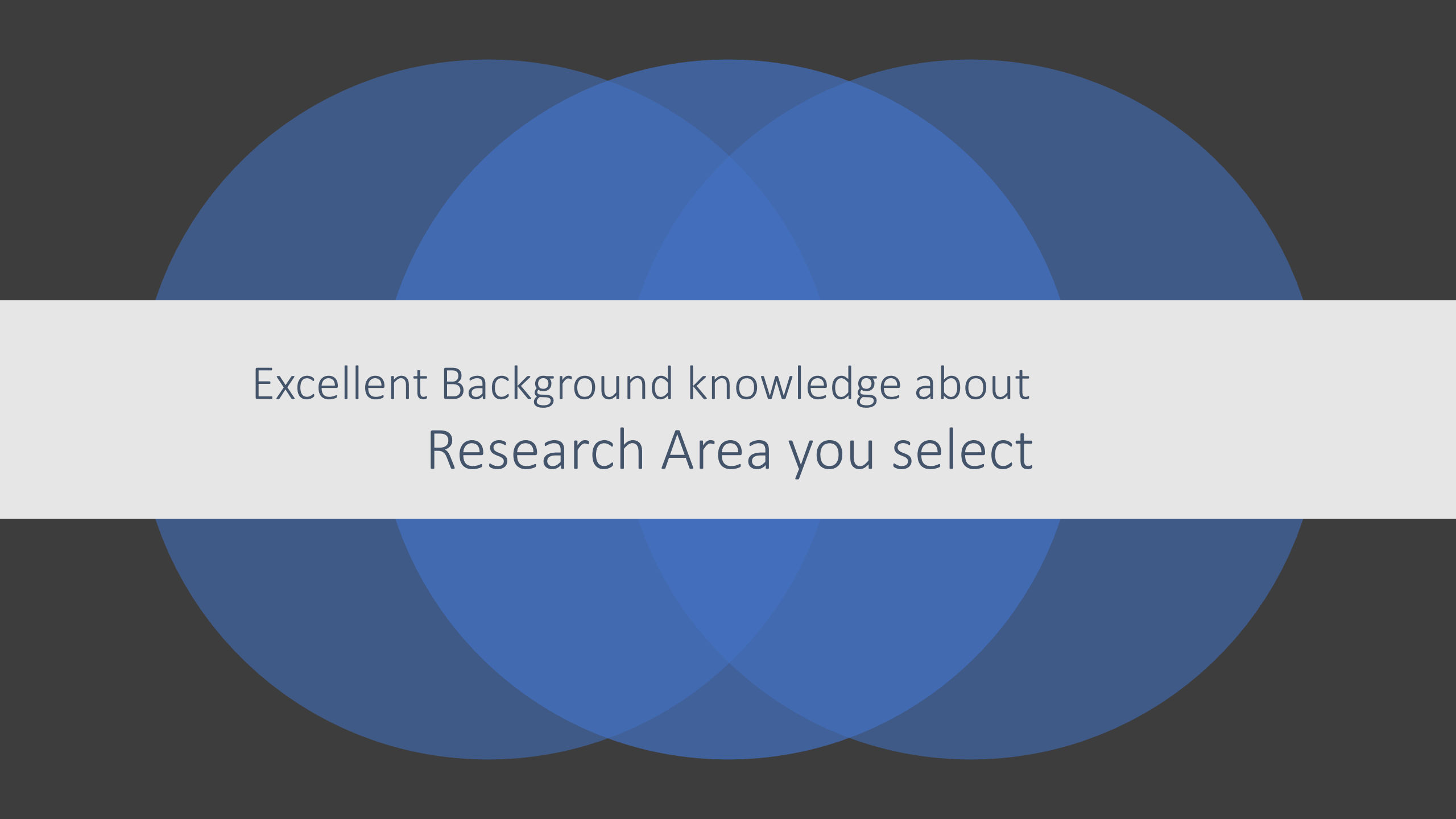


Machine Learning- Publication Analysis

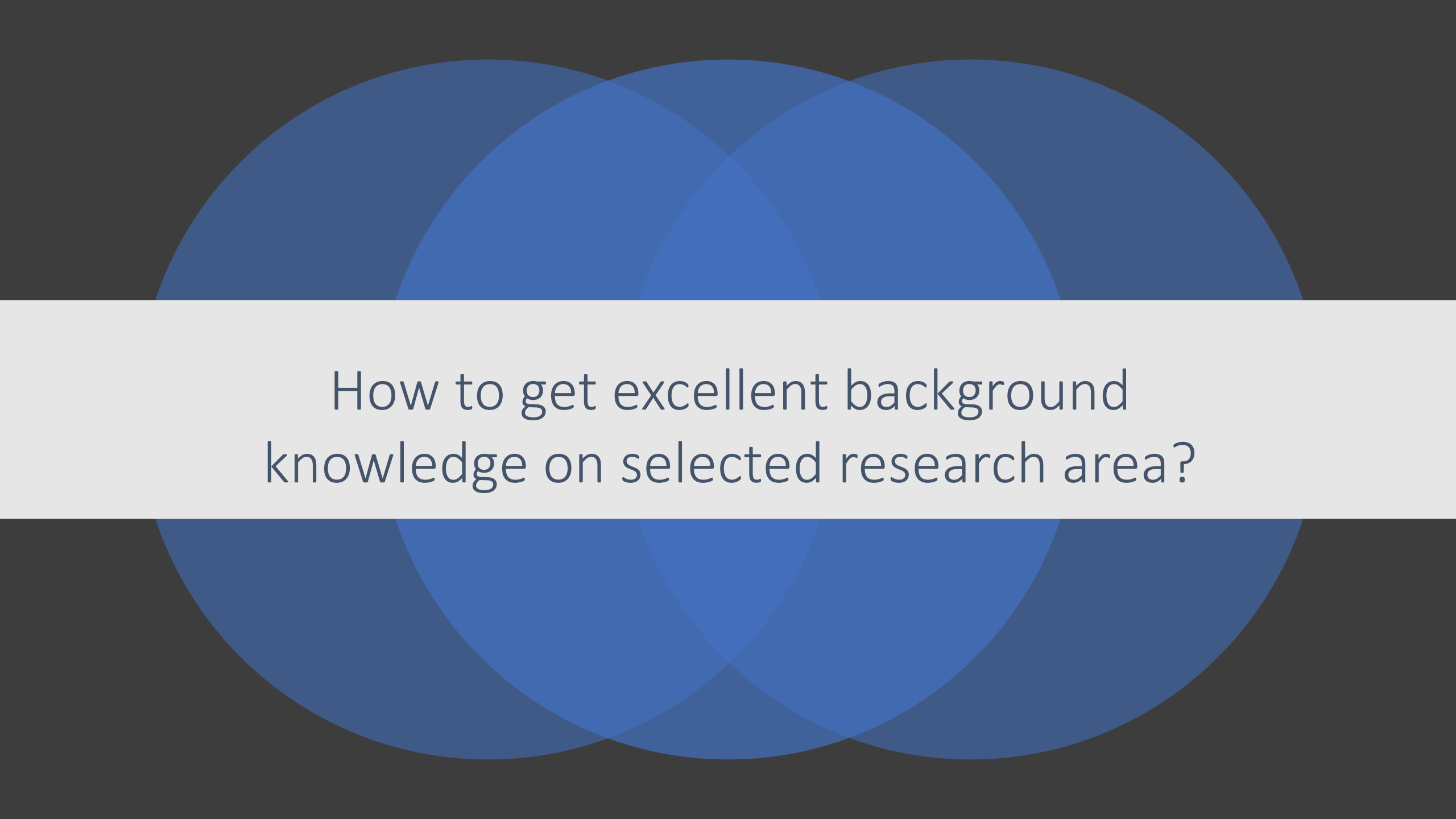


Machine Learning- Citations Analysis





Excellent Background knowledge about
Research Area you select



How to get excellent background
knowledge on selected research area?

How to get
excellent
background
knowledge on
selected
research
area?



Read iconic papers



Read book chapters



Watch video lectures



Listen talks of the experts

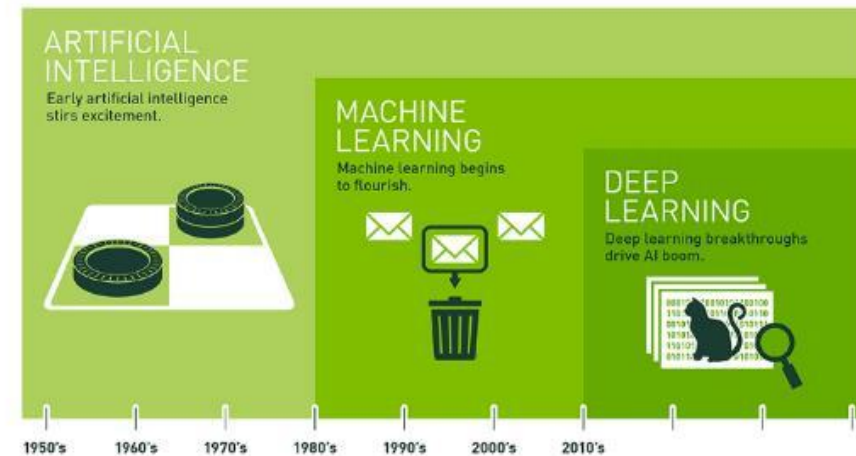
For Example- Deep Learning

23 Deep Learning Papers To Get You Started — Part 1



Rupak Kr. Thakur [Follow](#)

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Since an early flush of optimism in the 1950s, smaller subsets of artificial intelligence – first machine learning, then deep learning, a subset of machine learning – have created ever larger disruptions.

For Example- Deep Learning

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Any time

Since 2019

Since 2018

Since 2015

Custom range...

Sort by relevance

Sort by date

☒ include patents

☒ include citations

☒ Create alert

Deep learning

[Y LeCun, Y Bengio, G Hinton](#) - nature, 2015 - nature.com

Deep learning allows computational models that are composed of multiple processing layers to learn representations of data with multiple levels of abstraction. These methods have dramatically improved the state-of-the-art in speech recognition, visual object ...

  Cited by 18108 [Related articles](#) [All 53 versions](#)

[BOOK] Deep learning

[I Goodfellow, Y Bengio, A Courville](#) - 2016 - books.google.com

An introduction to a broad range of topics in deep learning, covering mathematical and conceptual background, deep learning techniques used in industry, and research perspectives. "Written by three experts in the field, Deep Learning is the only comprehensive ...

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Deep learning in neural networks: An overview

[J Schmidhuber](#) - Neural networks, 2015 - Elsevier

In recent years, **deep** artificial neural networks (including recurrent ones) have won numerous contests in pattern recognition and machine **learning**. This historical survey compactly summarizes relevant work, much of it from the previous millennium. Shallow and ...

  Cited by 6464 [Related articles](#) [All 35 versions](#)

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For Example- Deep Learning



Deep Learning In 5 Minutes | What Is Deep Learning? | Deep Learning Explained Simply | Simplilearn

Simplilearn • 48K views • 3 months ago

Don't forget to take the quiz at 04:26 Comment below what you think is the right answer, to be one of the 3 lucky winners who can ...



MIT Deep Learning Basics: Introduction and Overview

Lex Fridman • 357K views • 7 months ago

An introductory lecture for MIT course 6.S094 on the basics of deep learning including a few key ideas, subfields, and the big ...

CC



Seminário de Computação Gráfica: Deep learning: why all the hype? (2019)

Instituto de Matemática Pura e Aplicada • 822 views • 6 days ago

Speaker: Bianca Zadrozny - IBM Research - Brazil Abstract: In recent years, deep learning methods have been responsible for ...

New



MIT 6.S191 (2018): Introduction to Deep Learning

Alexander Amini • 417K views • 1 year ago

MIT Introduction to Deep Learning 6.S191: Lecture 1 Foundations of Deep Learning Lecturer: Alexander Amini January 2018 ...

For Example- Deep Learning



Applied Deep Learning with PyTorch - Full Course

freeCodeCamp.org 63K views • 7 months ago

In this course you will learn the key concepts behind deep learning and how to apply the concepts to a real-life project using ...

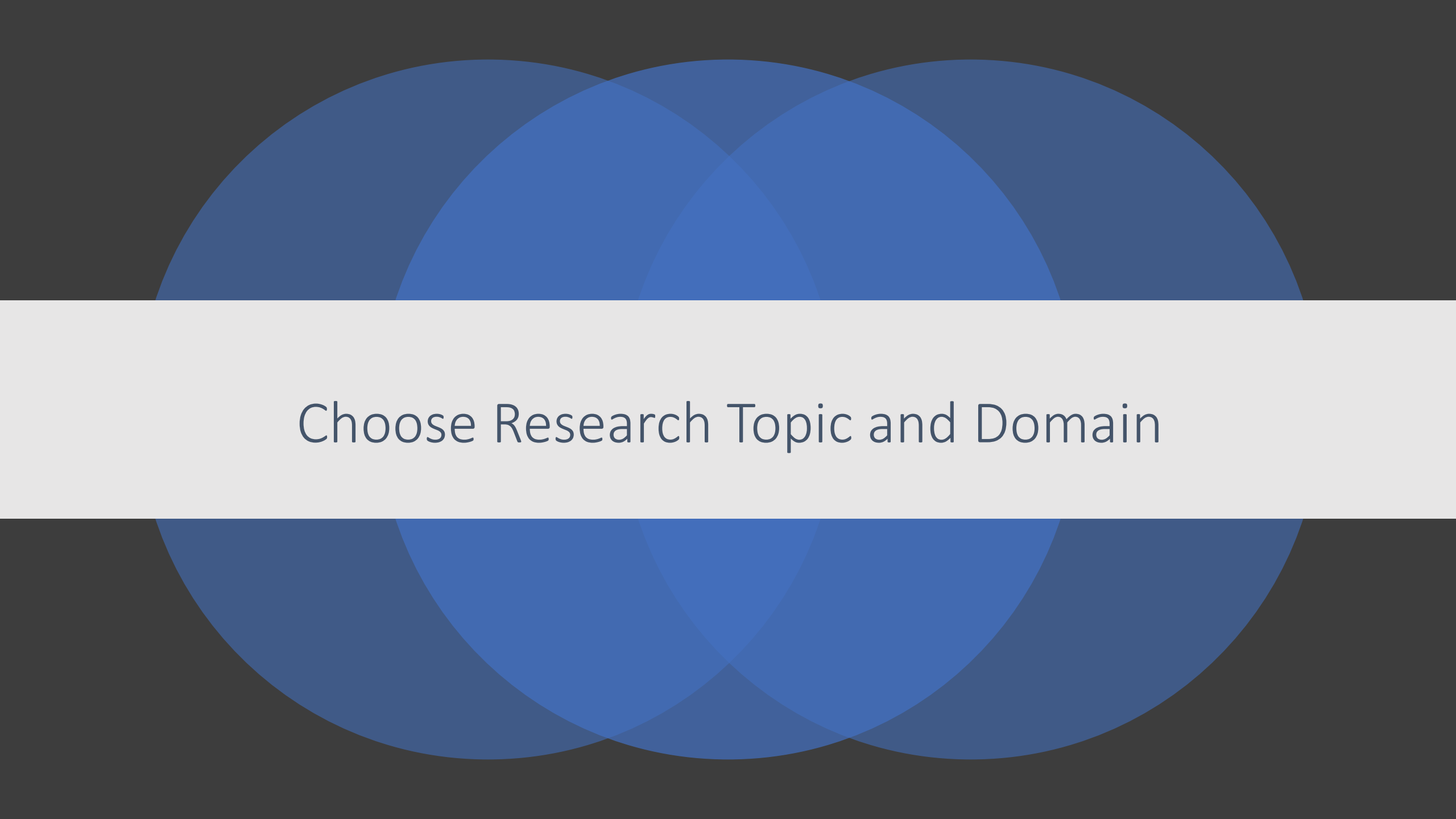


TensorFlow Full Course | Learn TensorFlow in 3 Hours | TensorFlow Tutorial For Beginners | Edureka

edureka! 64K views • 4 months ago

** TensorFlow Training: <https://www.edureka.co/ai-deep-learning-with-tensorflow> ** This Edureka TensorFlow Full Course video ...

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Choose Research Topic and Domain

Potential Research Topics in CS



Email Classification



Medical Reports Classification



Face Recognition



Image Processing



Image Classification



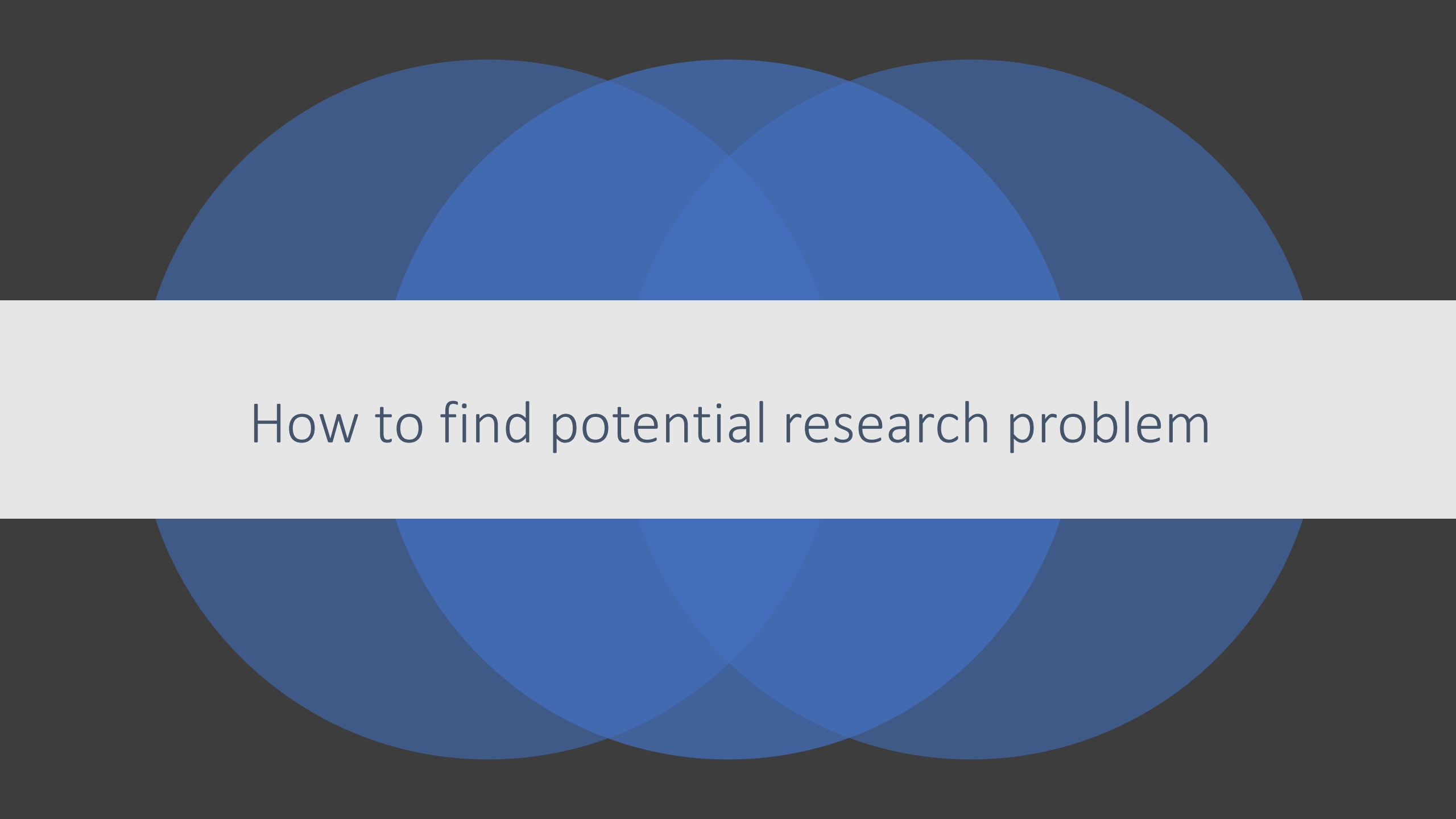
Medical Image Classification



Smart Cities



Internet of Things



How to find potential research problem

What is research problem?

1. To predict breast cancer from mammogram images. Is this a good research problem for CS postgrad students?
2. To classify email into ham or spam? Is this a good research problem for CS postgrad students?
3. To identify the car registration number using digital image processing. Is this a good research problem for CS postgrad students?

How to find potential research problem

- Extensive Literature review
- The pre-requisite for literature review is the extensive background knowledge about the research area.
- You must read some iconic paper followed by the latest research published in last 3-5 years.
- You must know how to read the literature or research paper
- Most of the time, the problem can be identified by reading the latest research articles on a specific research topic.

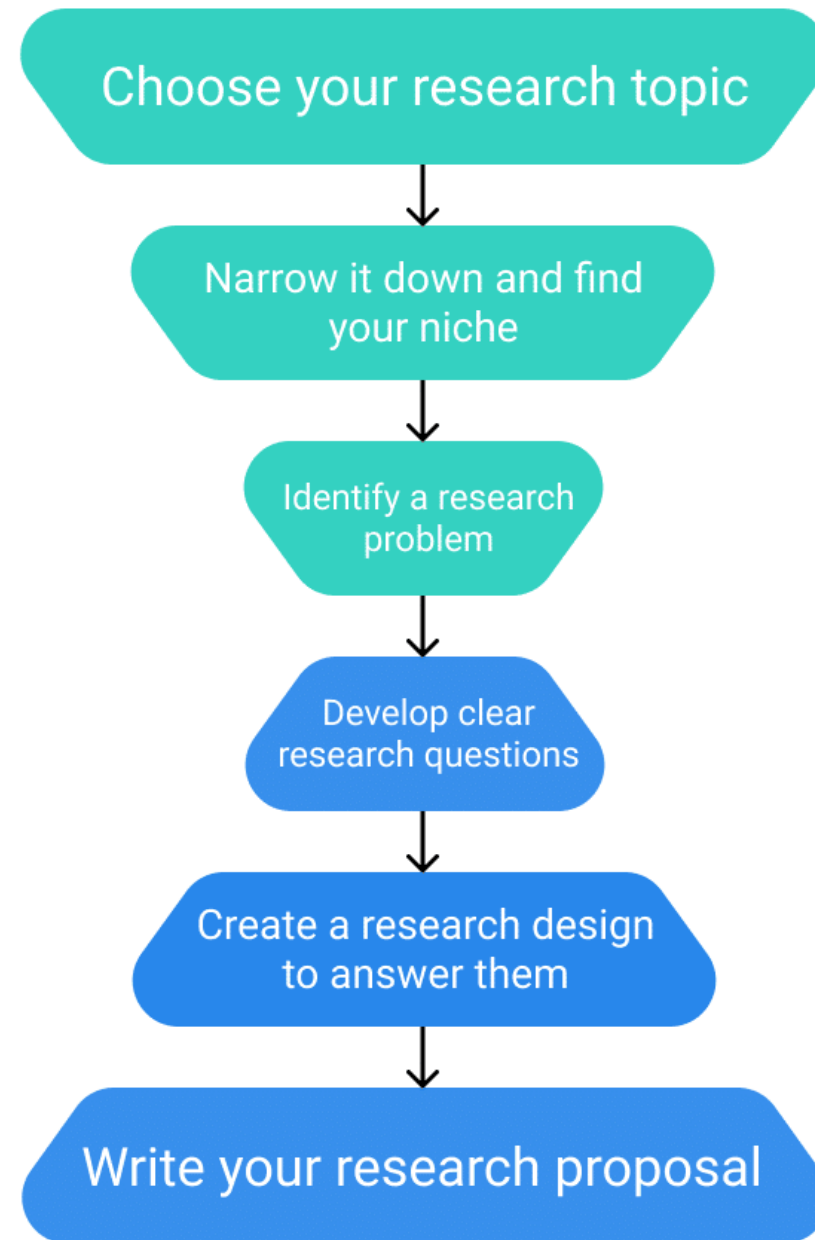
Example 1: Cybercrime detection in online communications: The experimental case of cyberbullying detection in the Twitter network

Example 2: Classification of forensic autopsy reports
through conceptual graph-based
document representation model



Research Proposal Development

Research Proposal Development Process



Recap

Basic Terminologies used in Research

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Types of Research according to use

Types of Research according to time

Types of Research according to data collection

Introducing a Research Process